CHECKPOINT: OPERATIONS, GOVERNANCE AND RISK MANAGEMENT

In this checkpoint we are aiming to acquire Digital Forensics Evidence

Investigating is a really important process in cyber security allowing us to find clues about the hackers gained access and the steps they followed in order to achieve persistence. Digital Forensics is a really important job allowing us to collect Evidence and analyse it and during this checkpoint you will learn how you can collect different data as Evidence from infected sources.

Giving instructions to follow from watching a video, from the video i will write a report:

Report: Acquiring Digital Forensics Evidence

1. Introduction

Digital forensics includes the collection, preservation, analysis, and presentation of digital evidence to aid investigations. In the context of compromised systems, this process is essential for determining how attackers accessed the system, the actions they took, and the full scope of the breach.

This report outlines the steps and tools used to collect evidence from infected sources.

2. Steps for Evidence Collection

a. Prepare the Forensics Environment

- Use dedicated forensic tools and an isolated system to analyze data.
- Ensure a write blocker is used to maintain the integrity of the evidence.
- Document the setup and maintain the chain of custody to track the evidence's handling.

b. Acquire Volatile Data

Volatile data is critical because it is lost when the system powers down. Use tools like Volatility or FTK Imager to collect:

- Running Processes: Identify suspicious processes using ps (Linux) or Task Manager (Windows).
- Open Network Connections: Capture active connections with netstat or Wireshark.
- System Memory (RAM): Dump the memory using tools like DumpIt or Belkasoft RAM Capturer.
- System Uptime and User Sessions: Determine login activity and remote sessions using commands like who or last.

c. Acquire Non-Volatile Data

Non-volatile data remains even after the system shuts down. Use disk imaging tools to preserve evidence.

- Disk Imaging: Create a bit-by-bit copy of the infected system using tools like dd (Linux), EnCase, or FTK Imager.
- System Logs: Extract logs from /var/log/ (Linux) or Event Viewer (Windows) for traces of unauthorized access.
- Registry Data (Windows): Investigate changes in the Windows Registry using Registry Viewer or Regshot.

d. Network Forensics

- Use tools like tcpdump or Wireshark to capture and analyze network traffic. Look for unusual traffic patterns, suspicious IP addresses, or command-and-control (C2) server communication.

e. Collecting Malware Samples

- Locate and isolate suspicious files or malware for analysis. Tools like Cuckoo Sandbox or VirusTotal can be used to safely analyze the behavior of the malware.

f. Snapshot the System

- Record the state of the infected system, including process trees, open ports, and scheduled tasks. Use commands like:
    - ps aux (Linux) or wmic process list (Windows) for process snapshots.
    - netstat -an for open network connections.
    - schtasks for scheduled tasks (Windows).

3. Tools for Evidence Collection

Below is a list of commonly used tools for different aspects of evidence collection:

- Memory Analysis: Volatility, DumpIt.
- Disk Imaging: FTK Imager, EnCase, dd.
- Log Analysis: Splunk, ELK Stack.
- Network Traffic Capture: Wireshark, tcpdump.
- Malware Analysis: Cuckoo Sandbox, VirusTotal.

4. Preservation and Chain of Custody

- Preserve Evidence Integrity: Use hashing (e.g., md5sum or sha256sum) to verify that evidence hasn't been altered.
- Secure Storage: Store evidence in tamper-proof containers or encrypted storage devices.

- Documentation: Record every step taken during the collection process, including timestamps, tools used, and actions performed.

5. Conclusion

Gathering digital forensics evidence is a detailed process that demands careful preparation, the use of specialized tools, and strict compliance with legal and procedural protocols. Adhering to best practices guarantees that the evidence is acceptable in legal proceedings and assists investigators in comprehending the scope and consequences of the incident.